

Biology 30 Unit A Sample Diploma-Style Exam 1

Biology 30 Unit A: Nervous and Endocrine Systems

Drawn from the attached Unit A lesson materials: Lesson 1 homeostasis and nervous-system foundations; Lesson 2 CNS integration; Lesson 3 sensory systems and perception; Lesson 4 endocrine regulation and nervous-endocrine integration.

Instructions: Select the best answer for each multiple-choice item. Answer constructed-response items using complete biological explanations. Each CR item is worth 3 marks unless otherwise directed.

Section 1 - Multiple Choice (60 questions)

MC 1. A student says homeostasis means internal conditions never change. Which correction is best?

- A. Homeostasis prevents all internal change from occurring.
- B. Homeostasis applies only to body temperature regulation.
- C. Homeostasis keeps internal variables within narrow limits despite ongoing fluctuations.
- D. Homeostasis occurs only when the body is at rest.

MC 2. A marathon runner has elevated body temperature after a race. Which component acts as the control centre in the thermoregulatory pathway?

- A. Sweat glands detect the temperature change and decide the response.
- B. Skin blood vessels generate the stimulus that begins the pathway.
- C. The hypothalamus compares temperature information with the normal range and coordinates responses.
- D. Skeletal muscles compare body temperature to the set point.

MC 3. In a negative feedback loop for thermoregulation, the response is best described as one that

- A. amplifies the original change until the process is completed.
- B. reduces the original change and returns the variable toward its normal range.
- C. prevents receptors from detecting the stimulus.
- D. occurs only through voluntary skeletal muscle action.

MC 4. If thermoreceptors cannot transmit signals to the hypothalamus, which outcome is most likely?

- A. Sweat glands will work normally because effectors do not need signals.
- B. Body temperature will be regulated more precisely by positive feedback.
- C. Effector responses may fail because the control centre lacks accurate input.
- D. The cerebrum will automatically replace all receptor function.

MC 5. Which example best identifies a receptor rather than an effector?

- A. A sweat gland secreting sweat.
- B. A skeletal muscle contracting during shivering.
- C. A blood vessel dilating near the skin surface.
- D. A skin thermoreceptor detecting increased temperature.

MC 6. Which statement best explains why the nervous system contributes to homeostasis?

- A. It prevents all endocrine glands from functioning.
- B. It rapidly detects changes and coordinates targeted responses.
- C. It maintains homeostasis only through conscious decisions.
- D. It replaces the need for receptors and effectors.

MC 7. Which sequence correctly describes information flow in many homeostatic pathways?

- A. Effector → stimulus → control centre → receptor → response.
- B. Control centre → stimulus → effector → receptor → response.
- C. Stimulus → receptor → control centre → effector → response.
- D. Receptor → effector → stimulus → control centre → response.

MC 8. A neuron at rest maintains unequal sodium and potassium ion concentrations across its membrane. Which process depends most directly on this condition?

- A. Release of hormones into the bloodstream.
- B. Maintenance of the resting potential.
- C. Replication of DNA in the nucleus.
- D. Digestion of neurotransmitters in the synaptic cleft.

MC 9. A drug blocks voltage-gated sodium channels in a sensory neuron. Which event would be impaired first?

- A. Diffusion of neurotransmitter across the synapse.
- B. Reuptake of neurotransmitter into vesicles.
- C. Secretion of insulin from pancreatic cells.
- D. Depolarization and action potential initiation.

MC 10. If a stronger stimulus is applied to a sensory neuron, how is stimulus strength usually represented?

- A. By a higher frequency or altered pattern of action potentials.
- B. By larger individual action potentials.
- C. By reversing the direction of impulse travel through axons.
- D. By converting electrical impulses into hormones.

MC 11. Which statement about a chemical synapse is most accurate?

- A. Neurotransmitters diffuse across the synaptic cleft and bind receptors on the postsynaptic cell.
- B. Neurotransmitters travel down the axon as electrical signals.
- C. Synapses allow cytoplasm to flow directly from one neuron to another.
- D. The postsynaptic cell always fires an action potential after any neurotransmitter binds.

MC 12. Increased inhibitory input to a postsynaptic neuron would most likely

- A. force the neuron to fire more frequently.
- B. make each action potential larger.
- C. turn the synapse into an endocrine gland.
- D. make the neuron less likely to reach threshold.

MC 13. Damage to myelin along a motor neuron would most directly affect

- A. the speed and coordination of impulse transmission.
- B. the genetic code used by the neuron.
- C. the ability of blood to carry hormones.
- D. the structure of pancreatic islets.

MC 14. A reflex protects the body mainly because it

- A. requires lengthy conscious analysis before movement.
- B. uses hormones instead of neurons.
- C. eliminates the need for sensory receptors.
- D. shortens response time to potentially harmful stimuli.

MC 15. Which pair correctly identifies the main divisions of the nervous system?

- A. Central nervous system and peripheral nervous system.
- B. Endocrine system and integumentary system.
- C. Cerebrum and pituitary system.
- D. Autonomic system and hormone system.

MC 16. Which structures make up the central nervous system?

- A. Brain and spinal cord.
- B. Brain and cranial nerves.
- C. Spinal cord and spinal nerves.
- D. Autonomic ganglia and endocrine glands.

MC 17. A person withdraws a hand from a sharp object before consciously identifying pain. Which explanation is best?

- A. The cerebrum initiates all protective responses first.
- B. The spinal cord can integrate some protective responses before conscious brain processing.
- C. Reflexes occur outside the central nervous system.
- D. Pain receptors send signals directly to skeletal muscle with no CNS involvement.

MC 18. Why is it incomplete to describe the spinal cord as only a wire carrying messages?

- A. It is part of the peripheral nervous system.
- B. It can integrate some information and produce rapid reflex responses.
- C. It produces all endocrine hormones.
- D. It is responsible for language comprehension.

MC 19. Damage to Broca's area would most directly impair

- A. visual image formation in the retina.
- B. blood glucose regulation by insulin.
- C. spinal reflex integration.
- D. fluent speech production.

MC 20. Damage to the occipital lobe would most directly affect

- A. voluntary hormone release from the pancreas.
- B. production of cerebrospinal fluid by bones.
- C. visual interpretation.
- D. chemical digestion in the stomach.

MC 21. The corpus callosum is best described as

- A. a gland that releases TSH.
- B. a sensory receptor in the retina.
- C. a fluid-filled space surrounding the spinal cord.
- D. a communication bridge between the cerebral hemispheres.

MC 22. Which lobe is most associated with planning, impulse control, and voluntary movement?

- A. Occipital lobe.
- B. Temporal lobe only.
- C. Frontal lobe.
- D. Cerebellar lobe.

MC 23. The cerebellum contributes most directly to

- A. secretion of insulin and glucagon.
- B. coordination, balance, and refinement of movement.
- C. production of action potentials at receptors.
- D. filtering urine in the nephron.

MC 24. The medulla oblongata is important because it helps regulate

- A. colour vision in the fovea.
- B. voluntary speech planning.
- C. long-term storage of all memories only.
- D. vital involuntary functions such as breathing and heart rate.

MC 25. Which statement best describes CNS integration?

- A. The CNS only forwards messages without processing.
- B. The CNS combines inputs, interprets them, and coordinates an appropriate response.
- C. The CNS acts only when endocrine glands are inactive.
- D. The CNS responds independently of receptors and effectors.

MC 26. Meninges, cerebrospinal fluid, and bone are important because they

- A. make nerve impulses travel in both directions.
- B. secrete all neurotransmitters in synapses.
- C. replace the need for myelin.
- D. help protect delicate central nervous system tissues.

MC 27. A patient can perform a withdrawal reflex but cannot learn from the painful event. This pattern most strongly suggests

- A. all CNS processing is absent.
- B. pain receptors are unnecessary for reflexes.
- C. endocrine glands are controlling the withdrawal reflex.
- D. spinal reflex pathways are intact but higher CNS processing is impaired.

MC 28. Which comparison of somatic and autonomic pathways is most accurate?

- A. Somatic pathways involve voluntary skeletal muscle control; autonomic pathways regulate involuntary effectors.
- B. Somatic pathways release hormones; autonomic pathways produce gametes.
- C. Somatic pathways are endocrine; autonomic pathways occur only in the cerebrum.
- D. Somatic pathways act on glands only; autonomic pathways act on skeletal muscle only.

MC 29. Sympathetic activation during a threat would most likely

- A. lower heart rate and promote digestion as the main response.
- B. block all sensory input from reaching the spinal cord.
- C. increase alertness and prepare the body for immediate action.
- D. convert blood glucose into bone minerals.

MC 30. Parasympathetic activity is most strongly associated with

- A. immediate fight-or-flight mobilization only.
- B. loss of all homeostatic feedback.
- C. voluntary contraction of biceps muscle only.
- D. restoration, digestion, and energy conservation.

MC 31. Why are sensory systems biologically necessary?

- A. They replace control centres in homeostatic systems.
- B. They generate responses without the CNS.
- C. They provide information needed to guide behaviour and homeostasis.
- D. They directly create perception without neural processing.

MC 32. A stimulus is best defined as

- A. only a change outside the body.
- B. only a voluntary action performed by muscles.
- C. a detectable internal or external change.
- D. a hormone released by any gland.

MC 33. Which evidence best shows receptors act as biological filters rather than perfect recorders of reality?

- A. Humans cannot detect ultraviolet light even though it exists.
- B. Humans see all wavelengths of electromagnetic radiation.
- C. The brain receives light directly from the environment.
- D. All receptors respond equally to every stimulus type.

MC 34. Which sequence best represents the general pattern shared by sensory systems?

- A. Stimulus → receptor → transduction → sensory pathway → brain interpretation.
- B. Perception → stimulus → receptor → sensory neuron.
- C. Stimulus → brain → receptor → action potential.
- D. Receptor → perception → stimulus → spinal cord.

MC 35. Why is it inaccurate to say the brain receives light or sound directly?

- A. Light and sound first become blood hormones.
- B. The spinal cord prevents all sensory input from reaching the brain.
- C. Photoreceptors and hair cells store stimuli permanently.
- D. The brain receives patterns of nerve impulses, not the original stimulus itself.

MC 36. A bright flash is perceived as more intense than a dim lamp mainly because

- A. the receptor response can lead to a higher frequency of impulses.
- B. each action potential from the flash is larger.
- C. light travels directly to the visual cortex.
- D. the thalamus bypasses the retina.

MC 37. Motion sickness most strongly supports which conclusion?

- A. Perception is a direct recording of reality.
- B. Balance depends only on visual input.
- C. The inner ear detects only chemicals in blood.
- D. Conflicting visual and vestibular input can alter perception and behaviour.

MC 38. Which statement best describes perception?

- A. Perception is a perfect copy of the external environment.
- B. Perception is an active interpretation of neural signals by the brain.
- C. Perception occurs only in sensory receptors.
- D. Perception does not depend on prior experience.

MC 39. The thalamus contributes to sensory processing by

- A. filtering, prioritizing, and distributing most sensory input.
- B. converting light into action potentials in the retina.
- C. secreting glucagon during fasting.
- D. acting as a passive wire that treats all input equally.

MC 40. Sensory adaptation is best described as

- A. a permanent loss of receptors after repeated use.
- B. increased response to every unchanging stimulus.
- C. reduced response to constant background stimuli so change is emphasized.
- D. a process that occurs only in the eye.

MC 41. Not noticing clothing pressure after several minutes is best explained by

- A. loss of all mechanoreceptors.
- B. sensory adaptation.
- C. damage to the occipital lobe.
- D. conversion of pressure into hormones.

MC 42. Damage to cone-rich foveal regions would most strongly disrupt

- A. balance controlled by semicircular canals.
- B. fine detail and reading ability.
- C. blood calcium regulation.
- D. sound amplification by ossicles.

MC 43. Hearing loss after cochlear hair-cell damage may be permanent because

- A. hair cells regenerate only in childhood.
- B. the tympanic membrane always blocks sound after damage.
- C. the auditory nerve becomes a visual fibre.
- D. hair cells in the organ of Corti do not regenerate effectively.

MC 44. A cochlear implant can partly restore hearing by

- A. regenerating all hair cells in the organ of Corti.
- B. bypassing damaged hair cells and stimulating auditory nerve fibres.
- C. changing sound waves into hormones.
- D. repairing the tympanic membrane automatically.

MC 45. Sensory overload would most directly result from impaired

- A. iris contraction only.
- B. lens accommodation only.
- C. thalamic filtering and prioritization.
- D. insulin secretion after a meal.

MC 46. A student argues that repeated nerve impulses could regulate blood glucose as effectively as hormones. Which response is best?

- A. Correct, because repeated impulses can replace all hormone function.
- B. Correct, because the pancreas is part of the nervous system.
- C. Incorrect, because glucose is regulated only by the kidneys.
- D. Incorrect, because blood glucose regulation requires sustained, multi-organ coordination better suited to hormones.

MC 47. Which scenario best demonstrates a problem requiring endocrine rather than primarily nervous control?

- A. Gradual coordination of puberty across many tissues.
- B. Immediate withdrawal from a pin.
- C. Rapid pupil constriction in bright light.
- D. Maintaining balance while walking.

MC 48. Which statement best captures the endocrine system as a time-based control system?

- A. It is always slower and therefore less important.
- B. It regulates processes that must remain stable over long periods.
- C. It specializes in reflexes and immediate muscle responses.
- D. It functions only when homeostasis fails.

MC 49. Why would 'faster is better' be poor reasoning in endocrine physiology?

- A. Many variables require gradual, stable adjustment rather than rapid all-or-none changes.
- B. Fast signaling prevents receptors from forming.
- C. Speed is useful only in skeletal muscle.
- D. Hormones are too weak to affect physiology.

MC 50. A hormone is present throughout the bloodstream, but only certain tissues respond. The best explanation is that

- A. only those tissues receive blood.
- B. the hormone chemically changes before reaching other tissues.
- C. only those tissues have the appropriate receptors.
- D. the pituitary blocks the hormone in other organs.

MC 51. Which finding best supports the idea that receptor presence matters more than hormone concentration alone?

- A. All cells respond equally when hormone levels are high.
- B. Hormones do not require target cells.
- C. Increasing hormone concentration has no effect in cells lacking receptors.
- D. Receptors determine which gland releases every hormone.

MC 52. A water-soluble hormone produces a response within seconds. Which mechanism is most consistent?

- A. It binds a surface receptor and triggers intracellular signaling.
- B. It diffuses into the nucleus and alters transcription directly.
- C. It acts without receptors because it dissolves in water.
- D. It permanently changes DNA sequence in target cells.

MC 53. A steroid hormone usually acts more slowly because it often

- A. travels through axons to target cells.
- B. requires no receptor on or in the cell.
- C. binds intracellular receptors and influences gene expression.
- D. is destroyed immediately in the bloodstream.

MC 54. Which statement best explains why endocrine disorders often produce multiple symptoms?

- A. Hormones circulate broadly and influence many tissues directly or indirectly.
- B. Each hormone affects only one receptor in one organ.
- C. Endocrine disease always spreads physically from gland to gland.
- D. Endocrine symptoms are psychological rather than physiological.

MC 55. Which case shows that endocrine disease is not simply too much or too little hormone?

- A. Low thyroxine reducing metabolism.
- B. Insulin lowering blood glucose.
- C. Receptor resistance causing poor response despite normal hormone levels.
- D. Glucagon increasing blood glucose during fasting.

MC 56. Negative feedback in endocrine pathways mainly helps

- A. guarantee positive feedback.
- B. fine-tune output and prevent overproduction.
- C. eliminate all baseline secretion.
- D. convert hormones into nerve impulses.

MC 57. During childbirth, oxytocin strengthens uterine contractions, which stimulate more oxytocin release. This is an example of

- A. negative feedback that immediately reduces contractions.
- B. positive feedback that ends when delivery removes the stimulus.
- C. sensory adaptation in the uterus.
- D. a spinal reflex unrelated to hormones.

MC 58. Why is the hypothalamus essential in endocrine regulation?

- A. It secretes all peripheral gland hormones directly into target cells.
- B. It prevents feedback from influencing endocrine glands.
- C. It is the only effector in blood glucose control.
- D. It links neural information with pituitary control of hormone release.

MC 59. Insulin and glucagon are both needed for homeostasis because they

- A. perform identical actions on blood glucose.
- B. coordinate opposite adjustments that keep blood glucose within a normal range.
- C. are both neurotransmitters released at synapses.
- D. act only during emergencies and not in daily life.

MC 60. Which comparison of epinephrine and cortisol is most consistent with stress physiology?

- A. Epinephrine supports rapid responses, while cortisol helps sustain longer stress effects.
- B. Both act only through intracellular receptors.
- C. Cortisol acts faster because it always changes gene expression instantly.
- D. Both are examples of purely voluntary nervous control.

Section 2 - Constructed Response (12 questions)

CR 1. A runner finishes a race on a hot day with elevated body temperature, sweating, and dilated skin blood vessels.

- Identify the stimulus in this homeostatic pathway.
- Describe the roles of the receptor, control centre, and effector.
- Explain why this is negative feedback rather than positive feedback.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 2. A toxin prevents thermoreceptors from sending impulses to the hypothalamus, although sweat glands are structurally normal.

- Predict the effect on temperature regulation.
- Explain why normal effectors may still fail to respond appropriately.
- Use this case to explain the importance of receptors in homeostasis.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 3. A student says reflexes are not part of the CNS because the brain is not consciously involved.

- State whether the claim is correct.
- Identify the CNS structure that integrates many reflexes.
- Explain why the brain still matters after the immediate reflex.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 4. A patient can withdraw from a painful stimulus but later cannot explain what happened or change future behaviour.

- Explain what this suggests about spinal reflex pathways.
- Explain what this suggests about higher CNS processing.
- Distinguish protective reflexes from learning.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 5. A bright camera flash and a dim lamp both activate photoreceptors, but the flash is perceived as more intense.

- Explain how both stimuli can be represented if action potentials are all-or-none.
- Explain the relationship between stimulus intensity and impulse frequency.
- Explain how this distinguishes receptor response from perception.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 6. A passenger reading in a moving car becomes nauseated.

- Identify the two sensory systems that are sending conflicting information.
- Explain why the conflict can lead to nausea.
- Explain what this shows about sensory integration.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 7. A patient detects sound normally in quiet conditions but cannot interpret speech well in a noisy room.

- Explain why this may not be a receptor problem.
- Explain how central processing and attention could contribute.
- Explain why sensory pathways are active rather than passive.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 8. A cochlear implant helps a patient hear after hair-cell damage but does not restore natural hearing perfectly.

- a. Describe how the technology works in general terms.
- b. Explain how it demonstrates neural plasticity.
- c. Explain why limitations remain.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 9. A student says the endocrine system is just a slower version of the nervous system.

- a. Explain why this statement is inaccurate.
- b. Identify one problem handled better by the nervous system and one handled better by the endocrine system.
- c. Explain why both systems are needed for survival.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 10. A hormone circulates throughout the bloodstream, but only liver, muscle, and adipose tissue respond.

- a. Explain why not all cells respond.
- b. Use the lock-and-key model to explain target specificity.
- c. Explain why broad circulation does not cause random effects.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 11. Blood glucose rises after a meal and later returns toward normal.

- a. Identify the main hormone released in response to rising blood glucose.
- b. Identify the major target tissues involved.
- c. Explain why this pathway is better suited to endocrine control than repeated nerve impulses.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 12. During labour, uterine contractions stimulate oxytocin release, and oxytocin strengthens contractions.

- a. Identify the type of feedback.
- b. Explain why the response amplifies the original change.
- c. Explain why the loop does not continue indefinitely.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

Section 3 - Answer Key and Scoring Guide

Multiple Choice Answer Key

1	C	16	A	31	C	46	D
2	C	17	B	32	C	47	A
3	B	18	B	33	A	48	B
4	C	19	D	34	A	49	A
5	D	20	C	35	D	50	C
6	B	21	D	36	A	51	C
7	C	22	C	37	D	52	A
8	B	23	B	38	B	53	C
9	D	24	D	39	A	54	A
10	A	25	B	40	C	55	C
11	A	26	D	41	B	56	B
12	D	27	D	42	B	57	B
13	A	28	A	43	D	58	D
14	D	29	C	44	B	59	B
15	A	30	D	45	C	60	A

MC Rationales

1. **C.** Homeostasis keeps internal variables within narrow limits despite ongoing fluctuations.
2. **C.** The hypothalamus compares temperature information with the normal range and coordinates responses.
3. **B.** reduces the original change and returns the variable toward its normal range.
4. **C.** Effector responses may fail because the control centre lacks accurate input.
5. **D.** A skin thermoreceptor detecting increased temperature.
6. **B.** It rapidly detects changes and coordinates targeted responses.
7. **C.** Stimulus → receptor → control centre → effector → response.
8. **B.** Maintenance of the resting potential.
9. **D.** Depolarization and action potential initiation.
10. **A.** By a higher frequency or altered pattern of action potentials.
11. **A.** Neurotransmitters diffuse across the synaptic cleft and bind receptors on the postsynaptic cell.
12. **D.** make the neuron less likely to reach threshold.
13. **A.** the speed and coordination of impulse transmission.
14. **D.** shortens response time to potentially harmful stimuli.
15. **A.** Central nervous system and peripheral nervous system.
16. **A.** Brain and spinal cord.
17. **B.** The spinal cord can integrate some protective responses before conscious brain processing.
18. **B.** It can integrate some information and produce rapid reflex responses.
19. **D.** fluent speech production.
20. **C.** visual interpretation.
21. **D.** a communication bridge between the cerebral hemispheres.
22. **C.** Frontal lobe.
23. **B.** coordination, balance, and refinement of movement.
24. **D.** vital involuntary functions such as breathing and heart rate.
25. **B.** The CNS combines inputs, interprets them, and coordinates an appropriate response.

- 26. **D.** help protect delicate central nervous system tissues.
- 27. **D.** spinal reflex pathways are intact but higher CNS processing is impaired.
- 28. **A.** Somatic pathways involve voluntary skeletal muscle control; autonomic pathways regulate involuntary effectors.
- 29. **C.** increase alertness and prepare the body for immediate action.
- 30. **D.** restoration, digestion, and energy conservation.
- 31. **C.** They provide information needed to guide behaviour and homeostasis.
- 32. **C.** a detectable internal or external change.
- 33. **A.** Humans cannot detect ultraviolet light even though it exists.
- 34. **A.** Stimulus → receptor → transduction → sensory pathway → brain interpretation.
- 35. **D.** The brain receives patterns of nerve impulses, not the original stimulus itself.
- 36. **A.** the receptor response can lead to a higher frequency of impulses.
- 37. **D.** Conflicting visual and vestibular input can alter perception and behaviour.
- 38. **B.** Perception is an active interpretation of neural signals by the brain.
- 39. **A.** filtering, prioritizing, and distributing most sensory input.
- 40. **C.** reduced response to constant background stimuli so change is emphasized.
- 41. **B.** sensory adaptation.
- 42. **B.** fine detail and reading ability.
- 43. **D.** hair cells in the organ of Corti do not regenerate effectively.
- 44. **B.** bypassing damaged hair cells and stimulating auditory nerve fibres.
- 45. **C.** thalamic filtering and prioritization.
- 46. **D.** Incorrect, because blood glucose regulation requires sustained, multi-organ coordination better suited to hormones.
- 47. **A.** Gradual coordination of puberty across many tissues.
- 48. **B.** It regulates processes that must remain stable over long periods.
- 49. **A.** Many variables require gradual, stable adjustment rather than rapid all-or-none changes.
- 50. **C.** only those tissues have the appropriate receptors.
- 51. **C.** Increasing hormone concentration has no effect in cells lacking receptors.
- 52. **A.** It binds a surface receptor and triggers intracellular signaling.
- 53. **C.** binds intracellular receptors and influences gene expression.
- 54. **A.** Hormones circulate broadly and influence many tissues directly or indirectly.
- 55. **C.** Receptor resistance causing poor response despite normal hormone levels.
- 56. **B.** fine-tune output and prevent overproduction.
- 57. **B.** positive feedback that ends when delivery removes the stimulus.
- 58. **D.** It links neural information with pituitary control of hormone release.
- 59. **B.** coordinate opposite adjustments that keep blood glucose within a normal range.
- 60. **A.** Epinephrine supports rapid responses, while cortisol helps sustain longer stress effects.

Constructed Response Model Answers and Rubrics

CR 1. A runner finishes a race on a hot day with elevated body temperature, sweating, and dilated skin blood vessels.

Model answer: a. The stimulus is increased body temperature above the normal range. b. Thermoreceptors detect the increase, the hypothalamus compares the input with the normal range and coordinates the response, and sweat glands plus skin blood vessels act as effectors that increase heat loss. c. The response is negative feedback because it reduces the original temperature increase and returns the variable toward normal.

Required concepts: Stimulus; thermoreceptors; hypothalamus; effectors; negative feedback; thermoregulation.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 2. A toxin prevents thermoreceptors from sending impulses to the hypothalamus, although sweat glands are structurally normal.

Model answer: a. Temperature regulation would be impaired because the control centre would not receive reliable information about body temperature. b. Sweat glands may be functional, but they need signals from the control centre; without receptor input, the hypothalamus may not activate them at the right time or intensity. c. Receptors begin the information flow by detecting the stimulus, so a homeostatic loop can fail even if the control centre and effectors are intact.

Required concepts: Receptor function; input to control centre; effector activation; homeostatic failure.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 3. A student says reflexes are not part of the CNS because the brain is not consciously involved.

Model answer: a. The claim is incorrect. b. Many reflexes are integrated in the spinal cord, which is part of the CNS. c. The brain receives information shortly afterward, allowing conscious awareness, interpretation, learning, and modification of future behaviour, even though conscious thought does not initiate the rapid withdrawal.

Required concepts: Spinal cord; CNS; reflex arc; conscious processing; learning.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 4. A patient can withdraw from a painful stimulus but later cannot explain what happened or change future behaviour.

Model answer: a. The intact withdrawal suggests spinal reflex pathways are functioning. b. The poor understanding and inability to use the event suggest impairment in higher brain processing involved in interpretation, memory, or decision-making. c. Reflexes provide rapid protection, while learning requires additional CNS processing that connects experience to future behaviour.

Required concepts: Reflex pathway; spinal integration; higher CNS processing; memory/decision-making.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 5. A bright camera flash and a dim lamp both activate photoreceptors, but the flash is perceived as more intense.

Model answer: a. Both stimuli are represented by patterns of neural activity rather than different-sized action potentials. b. A stronger stimulus usually produces a stronger receptor response and a higher frequency or different pattern of action potentials. c. Photoreceptors transduce light into neural signals, but conscious brightness is produced when the brain interprets the pattern.

Required concepts: All-or-none principle; frequency coding; transduction; perception.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 6. A passenger reading in a moving car becomes nauseated.

Model answer: a. The visual system and vestibular system are sending conflicting information. b. The eyes focused on a book may signal relative stillness, while the vestibular system detects motion; the mismatch can produce motion sickness and nausea. c. Perception depends on the brain integrating multiple sensory inputs, and conflict among inputs can disrupt perception and behaviour.

Required concepts: Visual system; vestibular system; multisensory integration; sensory conflict.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 7. A patient detects sound normally in quiet conditions but cannot interpret speech well in a noisy room.

Model answer: a. Detection can be normal while interpretation, filtering, or integration is impaired. b. Central processing, including thalamic filtering and cortical interpretation, helps prioritize relevant speech and reduce background noise. c. Sensory pathways actively modify, filter, and prioritize information before conscious perception.

Required concepts: Receptor vs central disorder; attention; thalamic filtering; cortical interpretation.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 8. A cochlear implant helps a patient hear after hair-cell damage but does not restore natural hearing perfectly.

Model answer: a. A cochlear implant bypasses damaged hair cells and stimulates auditory nerve fibres with electrical signals. b. It demonstrates plasticity because the brain can learn to interpret new patterns of input. c. Limitations remain because artificial stimulation does not reproduce the full precision, range, and adaptability of normal hair-cell transduction.

Required concepts: Cochlear implant; hair cells; auditory nerve; plasticity; limitations.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 9. A student says the endocrine system is just a slower version of the nervous system.

Model answer: a. The endocrine system is not simply slower; it solves long-term, gradual, system-wide regulation problems. b. The nervous system is better for immediate responses such as reflexes; the endocrine system is better for sustained processes such as blood glucose regulation, growth, metabolism, puberty, or prolonged stress. c. Survival requires both fast targeted responses and slower coordinated regulation across many organs.

Required concepts: Endocrine vs nervous control; timing; specificity; integration.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 10. A hormone circulates throughout the bloodstream, but only liver, muscle, and adipose tissue respond.

Model answer: a. Only cells with the correct receptor can respond. b. In the lock-and-key model, the hormone is the key and the receptor is the lock; only a matching receptor can bind the hormone and initiate signaling. c. The hormone may reach many cells, but cells without receptors ignore the message, so effects are selective.

Required concepts: Hormone receptor; target cell; lock-and-key; specificity.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 11. Blood glucose rises after a meal and later returns toward normal.

Model answer: a. Insulin is released by pancreatic beta cells when blood glucose rises. b. Liver, skeletal muscle, and adipose tissue are major target tissues. c. Hormones are suited because glucose regulation requires sustained, multi-organ coordination over time rather than brief localized impulses.

Required concepts: Insulin; pancreas; liver/muscle/adipose tissue; sustained endocrine coordination.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 12. During labour, uterine contractions stimulate oxytocin release, and oxytocin strengthens contractions.

Model answer: a. This is positive feedback. b. Contractions stimulate oxytocin release, and oxytocin causes stronger contractions, increasing the original response. c. The loop ends when delivery removes the stimulus that maintained the cycle.

Required concepts: Positive feedback; oxytocin; childbirth; loop termination.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.